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To: Hannah Thompson, Planning Plus Ltd

From: Dr Jamie MacKay

Date: 5 July 2018

Subject: **APPLICATION NUMBER SUB60315119 (OUR REF: 4384V)**

INTRODUCTION

Ray and Desma Nicklin ("the applicants") are applying to subdivide a rural property at 53B Remiger Road, Puhoi. The subdivision is being applied for under Rule 7.14.3.2 of the Auckland District Plan (Rodney Section) whereby a property may be subdivided in return for the protection of at least two hectares of significant indigenous bush. The applicants propose to protect 53.43 hectares of indigenous vegetation to enable the creation of five additional lots.

Auckland Council commissioned Wildland Consultants Ltd to undertake a site visit and to provide a review of the ecological reporting for the resource consent application. This report outlines the findings of our review of the ecological information provided. I was initially provided with the following reports, which form the basis of my assessment:

- Resource Consent Application. Cato Bolam Ltd, 30 November 2017.
- Bush Quality Assessment Report. Cato Bolam Ltd, 1 September 2017.
- Riparian Revegetation and Management Report. Cato Bolam Ltd, 1 October 2017.
- Scheme Plan. Cato Bolam Ltd, September 2017.

I visited the site on 8 March 2018 and spent approximately four hours assessing the vegetation within the proposed bush protection areas.

SECTION 92 REQUEST FOR FURTHER INFORMATION

Following my site visit I submitted a Section 92 request for further information regarding a watercourse that flows through proposed new Lots 3, 4, and 5. The applicants provided a response to the S92 request dated 23 March 2018.

REVIEW OF THE BUSH ASSESSMENT REPORT

Summary of the applicant's report

In order to qualify for an additional lot under Rule 7.14.3.2 of the Rodney District Plan, at least two hectares of good quality indigenous bush must be present on the property. Under the District Plan rules, between 42 and 52 hectares of indigenous bush must be protected to qualify for five additional lots. The quality of the proposed bush lots are assessed against the guidelines outlined in Appendix 7B of the plan. Two scales are used for assessment: the first (Scale 1) focuses on bush quality and the second (Scale 2) on other features. A high score on Scale 1

indicates that the bush is significant and that Scale 2 is unnecessary. A stand of bush that scores marginally on Scale 1 may be suitable for protection if it scores well in the Scale 2 assessment.

Four stands of significant indigenous bush with a combined area of 53.43 hectares are proposed for protection. Three bush quality assessments were undertaken in three of the proposed covenants at locations considered representative of vegetation within the proposed covenant. No bush quality assessment was undertaken in Area CW. This is the smallest of the proposed covenants (1.4 hectares) and the applicant's report states that it is closely associated with Area CV. It is thus assumed that due to its small size and similarity to Area CV, no bush quality assessment was undertaken in Area CW.

The following table summarises the bush assessment scores reported for the four proposed bush covenants.

Covenant Area	Area	Scale 1 Average Score	Scale 2 Average Score	Comment
CU, CL, CM, CN, CO, CP, CQ, CR	37.2 hectares	20	15	SEA, qualifies without Scale 2
CS, CT	8.1 hectares	19	5	Non-SEA, relies on Scale 2
CV	6.7 hectares	24	0	Non-SEA, qualifies without Scale 2
CW	1.4 hectares	n/a	n/a	Non-SEA, no bush quality assessment undertaken

The report concludes that each of the indigenous bush areas that were assessed qualify as significant bush under the guidelines outlined in Appendix 7B of the plan.

Review of the applicant's report and assessment under the Rodney District Plan

I visited each of the proposed bush covenants where bush assessments were undertaken during my site visit. I agree with the scores assigned to the bush areas, and the site descriptions provided in the applicant's assessment concur with my observations. The proposed bush covenants do qualify as significant under Rule 7.14.3.2 of the Rodney District Plan. Protection of the bush areas through fencing, cessation of grazing, ongoing pest plant and animal control, and permanent legal protection will result in increased ecological benefits for the property and the surrounding area.

Assessment under the Auckland Unitary Plan

In order to qualify for subdivision under the Auckland Unitary Plan (AUP), proposed bush covenants must be within a Significant Ecological Area (SEA) overlay. Fencing around the proposed covenants must be 20 metres away from the bush edge to provide a protective buffer to the covenanted vegetation. In addition, only three new in-situ titles are permitted under the AUP. The proposal does not qualify for subdivision under the AUP in its current form.

Summary

Vegetation within the proposed indigenous bush covenants at the property meets the criteria to be classed as significant under Rule 7.14.3.2 of the Rodney District Plan. If the application is assessed under the Rodney District Plan, I recommend that consent to subdivide be granted subject to the conditions outlined below.

The current application proposes to protect one area of bush within an SEA overlay. Fencing for this proposed covenant will be installed at the bush edge rather than 20 metres back from the bush edge, and as such the application does not qualify for subdivision under the AUP rules. There is sufficient indigenous vegetation within an SEA overlay present at the property to allow subdivision into several lots consisting of a combination of in situ titles and transferrable titles if the fencing requirements are met. Protection of the bush areas that are not within the SEA overlay will not qualify for subdivision under the AUP.

REVIEW OF THE RIPARIAN REVEGETATION AND MANAGEMENT REPORT

The Riparian Revegetation and Management Report contains guidelines on fencing, site preparation, pest plant and animal control measures, and plant schedules for the proposed riparian planting areas on proposed Lots 4 and 5. The report refers to a Weed and Pest Management Report which I have not been provided, although I understand it has been reviewed by Auckland Council.

Fencing

The proposed permanent seven-wire post and batten fence is acceptable under Rule 7.14.2.5.2 of the Rodney District Plan.

Pest plant control

Pest plant control measures are outlined in the Weed and Pest Management Report, which I have not reviewed. The presence of kikuyu is noted and the importance of regular ongoing control of this species to prevent smothering of the plantings is emphasised.

Pest animal control

As with pest plant control, the main pest animal control recommendations are included in the Weed and Pest Management Report. The Riparian Revegetation and Management Report provides guidelines for the control of rabbits, hares, and pūkeko (if required). The methods outlined in the plan are appropriate; however, it may be necessary to shoot rabbits and hares as poisoning can sometimes be ineffective.

Planting

The planting guidelines are appropriate for the site. The plant schedule is generally acceptable, although some changes are recommended. Firstly, rewarewa and pūriri are considered inappropriate for the site. Rewarewa does not grow well in wet situations and it is likely that pūriri will be damaged by frosts if planted in the open, as appears to be the case here. It is recommended that these species are substituted for 45 kahikatea. Secondly, the suggested plant grades for tōtara and kahikatea are too small and these should be increased to a minimum size of two litre grades. Thirdly, a note should be made that harakeke/flax should not be planted on the edge of the watercourses as it is susceptible to being dislodged during floods. The recommended plant spacings are appropriate for the site.

Summary

Fencing, pest plant and animal guidelines, and planting guidelines are appropriate. The plant schedule should be updated as per the table below:

Species	Grade	Quantity	Note
Mānuka	0.5 litre	2,500	
Kānuka	0.5 litre	1,750	
Karamu		800	
Cabbage tree		800	
Harakeke/NZ flax		255	Should not be planted on the edge of the watercourse
Tōtara	2 litre	50	Should not be planted on the edge of the watercourse
Pūriri	n/a	0	Should not be planted due to susceptibility to frost
Rewarewa	n/a	0	Should not be planted in riparian areas due to intolerance of wet soils
Kahikatea	2 litre	45	Can be planted in very wet areas

Implementing this plan in full will improve the riparian ecological values at the property.

WATERCOURSE CLASSIFICATION

A small watercourse flows through proposed Lots 3, 4, and 5.
The scheme plan identifies the lower reach of the watercourse within proposed Lot 5 as requiring riparian planting; however, riparian planting is not proposed for the rest of the watercourse. I queried the watercourse status in the S92 request. The applicant's response included a watercourse assessment under the AUP that concluded that the watercourse was ephemeral rather than intermittent.

Auckland Council guidelines for the distinction between ephemeral and intermittent watercourses require that the watercourses are assessed against several criteria relating to channel morphology, hydrology, vegetation, and substrate. These aspects were covered in the applicant's S92 response and my assessment is outlined below.

Channel morphology

The applicant's assessment concluded that "*the watercourse is more akin to a wetland environment as the wet area is spread over a much greater width than a simple stream channel*". Whilst I agree that watercourse has wetland characteristics, I do not agree that this precludes it being classified as intermittent.

The upper reach of the watercourse on proposed Lot 3 is severely degraded through cattle pugging and through a partially blocked culvert underneath a driveway, which in turn has resulted in an induced wetland habitat. Despite this, evidence of the historic channel morphology is still visible. Stock are able to access the watercourse downstream of the driveway where it flows through proposed Lots 4 and 5; however, the flow is not restricted here and the watercourse has a more discernible defined channel. The presence of a defined channel is a key indicator of an intermittent watercourse as per the criteria in the AUP.

Hydrology

My assessment took place during a period of heavy rain and it was therefore not possible to accurately assess the hydrology of the watercourse. The criterion to be met as outlined in the applicant's report was "*it contains surface water more than 48 hours after a rain event which results in stream flow*". The applicant's report states that surface water but no flow was present thus indicating the watercourse is ephemeral. A more detailed description of this criterion is provided in a draft working document prepared by Auckland Council entitled *Guidance for the classification of drainage systems based on the Proposed Auckland Unitary Plan definition.*

This document highlights the constraints associated in assessing a watercourse 48 hours after the previous rainfall event, including the fact that in order to fully satisfy the wording of the criterion the watercourse must be assessed at exactly 48 hours after the last rainfall event. Further conditions that must be satisfied whilst assessing this criterion including the amount of rainfall in the previous 24 hours (a minimum of 10 millimetres) and whether or not the rainfall event produced runoff resulting in stream flow are also discussed. The fact that it was raining during the assessment mean it was not possible to use this criterion during this assessment.

The presence of natural pools is also considered an indicator of an intermittent rather than ephemeral watercourse; however, an assessment of this criterion is not included in the draft guidance document.

Vegetation

The presence or absence of rooted terrestrial vegetation indirectly indicates the length of time the watercourse channel contains water. This criterion can be challenging to assess because certain terrestrial plant species (particularly pasture species) can persist for prolonged periods in waterlogged or inundated conditions and also colonise rapidly as habitats dry out. For this reason terrestrial vegetation is likely to dominate intermittent watercourses during low-flow conditions through spring and summer. The presence of obligate or facultative wetland plant species[1] is an indication that the watercourse is more likely to be intermittent than ephemeral.

The applicant's report states that terrestrial vegetation is not established across the entire cross-sectional width of the channel, which is a characteristic of intermittent watercourses. I agree with this assessment.

Substrate

Intermittent watercourses deposit organic debris on floodplains during flood events and substrates become sorted (different sized substrate particles become aggregated in relation to flow rates). The applicant's report states that there is no evidence of either process occurring. I did not make a full assessment of either process during my site visit as I considered the channel morphology and vegetation characteristics of the watercourse were sufficient to classify the watercourse as intermittent.

Summary

I consider the watercourse meets the criteria to be classified as intermittent. This means that riparian planting at a width of ten metres on each side of the watercourse is required along its entire length. In this respect, my conclusion differs from the one reached by the applicant's ecologist. Accordingly, I suggest that the watercourse is assessed by an Auckland Council specialist at least 48 hours after the last rainfall event in order to provide a third opinion of the status of the watercourse. If the watercourse is determined to be intermittent rather than ephemeral a riparian planting plan will be required for the entire watercourse.

RECOMMENDED CONSENT CONDITIONS

The following conditions of consent are recommended:

[1] Clarkson BR, Champion PD, Rance BD, Johnson PN, Bodmin KA, Forester L, Gerbeaux P, Reeves PN 2013. Wetland indicator status ratings for New Zealand species. Landcare Research, Hamilton http://www.landcareresearch.co.nz/__data/assets/pdf_file/0014/64400/wetland_rating_species_December_2013.pdf.

- All fences shall be maintained in perpetuity to ensure the continued exclusion of livestock from all covenanted areas.
- The restoration methods outlined in the Riparian Revegetation and Management Report shall be fully implemented. The plant schedule within the report shall be updated to remove rewarewa and pūriri, and the grades of tōtara and kahikatea shall be increased to 2 litres.
- The fenced areas shall be legally protected in perpetuity under a QEII Natural Trust open space covenant or similar mechanism.
- A consent notice shall be established on each title to ensure that property owners are aware of their obligations in respect of the covenanted vegetation on their land.

CONCLUSION

The applicant has proposed to subdivide a rural property at 53B Remiger Road, Puhoi, in order to create five new in situ lots in return for the protection of significant indigenous vegetation. The application has been applied for under the Rodney District Plan rather than the Auckland Unitary Plan.

The proposed bush covenants meet the significance criteria under the Rodney District plan. The proposed riparian restoration methodologies are acceptable provided the plant schedule is updated as per the recommendations above. I have not reviewed the Weed and Pest Management Plan prepared for the property.

In my opinion a watercourse that flows through proposed new Lots 3, 4, and 5 meets the criteria to be classified as intermittent under the AUP; however, the applicant states that it is ephemeral. I suggest that a third opinion be sought from an Auckland Council specialist before consent is granted. If the watercourse is determined to be intermittent a ten metre riparian margin must be fenced and restored on each side of the watercourse. The current approach for riparian restoration is appropriate for the watercourse reaches on proposed new Lots 4 and 5. The reach on Lot 3 has been degraded by stock and possibly through a partially blocked culvert underneath a driveway, which in turn has resulted in an induced wetland habitat. This habitat will require a different plant schedule, which includes indigenous wetland sedge and rush species.

In conclusion, I recommend that consent to subdivide the property is granted based on the consent conditions listed above. A decision must be reached on the status of the watercourse before a final scheme plan can be approved. If the watercourse is determined to be intermittent an updated Riparian Revegetation and Management report will need to be provided.



Dr Jamie MacKay
Senior Ecologist

Hannah Thomson

From: Jamie MacKay <Jamie.MacKay@wildlands.co.nz>
Sent: Friday, 9 March 2018 9:00 AM
To: Hannah Thomson
Subject: RE: Brief- SUB60315119- 53a Remiger Road

Hi Hannah,

I visited the site yesterday and had a good look round. It is a bit of a complicated one in terms of the different plans – I note that throughout all the Cato reports they constantly refer to the "still operative" Rodney plan but in our recent experience with another site (where the consent application was originally submitted long before the new rules came in) that isn't true. My first thoughts on the assessment under the two plans:

Rodney plan

- The bush blocks all qualify so no problems there
- There is an intermittent to permanent watercourse that starts in Lot 3 and flows through Lots 4 and 5. The very bottom reach of this watercourse is proposed for planting (area Z on the scheme plan) but the rest isn't and it should be. To phrase this as an S92 request:
    - No fencing and riparian planting is proposed for a watercourse identified on the Auckland Council GIS system flowing through Lots 3, 4 and 5. This watercourse is degraded; however, it has a defined channel and meets the criteria to be determined as Intermittent under the AUP. Please provide an updated scheme plan showing fencing and riparian planting around this watercourse.

AUP

- Proposed covenant CU, CP, CO and CR are within the SEA overlay so protection of these area with a 20 metre buffer qualifies for subdivision
- Proposed covenants CT, CS, CV and CW are not SEA so protection of these areas does not qualify for subdivision. All of these should be SEA
- A block of SEA vegetation at the northern section of the site does qualify for subdivision but that isn't going to be protected under this proposal.

In this case I think assessing the application under the Rodney plan will create a better ecological outcome because a larger area of indigenous forest will be protected. I expect that Cato are going to have some sort of argument against protecting the unprotected watercourse but that could be used as a negotiating tool to get the application assessed under Rodney rather than AUP but I don't know if it works like that. Another option could be to get the better quality patches of bush classed as SEA but I don't think this an option anymore. I would support a compromise where these blocks are assumed to be SEA and protected with a 20 metre buffer if that gets the applicant the outcome he wants but if we push through AUP rules he might walk away from the non-SEA blocks of bush. This may result in the other SEA bush being protected but then the smaller patches won't be.

So...a bit of a tricky one. It would be good to have a chat with you about it before I put too much time into the report. I am in the office all day if you are free for a call.

Cheers,

Jamie

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From: Hannah Thomson [mailto:hannah@planningplus.co.nz]
Sent: Wednesday, 7 March 2018 12:46 p.m.
To: Jamie MacKay
Subject: RE: Brief- SUB60315119- 53a Remiger Road

ENGINEERING APPRAISAL MEMORANDUM **5/03/2018**

To: Hannah Thomson

From: **Ronald Uy**
(Peer reviewed by Vijay Kunaseelan)

Application Number(s):	SUB60315119
Applicant:	Raymond Charles Nicklin
Proposed Activity(s):	Rural subdivision to establish 5 freehold rural-residential allotments on the basis of protecting significant native bush.
Address:	53B Remiger Road, Puhoi

As requested, I have assessed the application against Council's relevant engineering policies. These include:

- **Access**

Access to the proposed lots via the existing right of way / vehicle crossing branching off from the existing metaled Remiger Road. On-site parking.

- **Earthworks**

Earthworks for the creation / upgrade of the new ROW anticipated not to exceed the permitted activity thresholds.

- **Geotechnical Issues**

Category B soils – slightly expansive. Geotech report identified category M soils in terms of AS2870:2011. Site away from a northern allochthon overlay. Proposed building platform sites ranging from moderately sloping (1:5.5) to relatively flat. Geotechnical Investigation Report prepared by Engineering Geology Ltd., ref: 8447, dated 14 September 2017.

- **Wastewater**

On-site effluent treatment and disposal system.

- **Stormwater**

Roof runoff from dwellings on lots 1-5 will flow to reuse tanks and overflow will be piped for sheetflow discharge over pasture. Driveway / paving runoff on lots 1-5 will bypass tank system and piped to a level spreader dispersal device.

Flooding and Overland Flowpaths

Council Geomaps indicated overland flow paths and floodplains within the property but the proposed building platforms are away from them. The vehicle crossing and the first 70 meters (roughly) of the

ROW however are within the flood plains overlay. Site is away from coastal inundation.

- **Water Supply**

Roof rain runoff reuse tanks.

- **Services**

Power and telephone connections will be required for each lot.

- **Other matters**

As part of this assessment, a site visit to the subject site was carried out on 26 March 2018.

Having assessed the application, I recommend that, subject to the following conditions the consent can be granted from an engineering perspective.

Engineering Conditions - To be completed before issue of the s.224(c) certificate

(construction of right of way) The right of way over parts of proposed Lot 6 up to the intersection at Remiger Road shall be upgraded to a rural metaled standard to the Council's "Standards for Engineering Design and Construction". Sections with a grade greater than 1 in 8 shall be to a rural concrete standard. The upgrade shall ensure that the formed carriageway is no less than 3 metres in width. Further it shall include strategically placed passing bay/s for the purpose of servicing lots and make adequate provision for the drainage of surface water as well as provision of any ducts required for power and communications.

(engineering plans) The engineering works required by this consent shall comply with the Council's "Standards for Engineering Design and Construction" as may be amended from time to time. Engineering Plans, as specified in the "Standards", shall be submitted to the Consents Engineer, and approval thereto received in writing, prior to the commencement of any works on the site.

Any variation or changes to the approved engineering plans shall be submitted for approval as an Amendment and approval received thereto prior to construction of the varied works.
The term 'engineering works' includes, but is not limited to:

1.1 Earthworks,

1.2 Any other works required by conditions of this consent.

Note: Structures such as retaining walls, in-ground walls and bridges will require a separate Building Consent.

Note: The plans required under this condition are separate to, and do not form part of, any Building Consent that may be required on the subject site.

(silt retention) Before commencement of any works, adequate silt retention structures as detailed in Technical Publication No. 90 "Erosion and Sediment Control Guidelines for Land Disturbing Activities in the Auckland Region" shall be installed. These structures shall be maintained and cleaned out as necessary until such time as complete grass cover, or other non-erodible surface has been re-established over the area. The location and detail of the sediment control shall be included in the Engineering Plans for approval prior to commencement of construction.

(provide for electric power) Written confirmation shall be provided from the electricity network supplier responsible for the area, that provision of an electric supply has been made available by underground means to all saleable lots created and that all the network supplier's requirements for making such means of supply available have been met or satisfactory arrangements have been concluded with the Consent Holder to complete the provision of the supply.

(provide for telecommunications) Written confirmation shall be provided from the telecommunications network supplier responsible for the area, that provision of telephone services has been made available by underground means to all saleable lots created and that all the network supplier's requirements for making such services available have been met or satisfactory arrangements have been concluded with the Consent Holder to complete the provision of the service.

(new road name) The Consent Holder shall suggest to the Council a name for the new road shown on the Scheme Plan after consulting the local iwi for comment, together with clearance from Addresses@linz.govt.nz so that duplication of the name in any other part of the Auckland region is avoided. The Council Surveyor should be consulted in the first instance in regard to the appropriateness of the name and road type. When a name has been resolved with Local Board approval, the Consent Holder shall erect nameplates, in accordance with the Council's "Standards for Engineering Design and Construction".

Ongoing Conditions/Consent Notices

a) (firefighting supply) Upon the construction of habitable buildings, sufficient water volume shall be provided in accordance with NZFS Fire Fighting Water Supplies CoP SNZ 4509:2008 and that this water supply be accessible for firefighting purposes. Should the water supply be provided by way of tank storage, this storage must be located a safe distance away from any habitable dwelling in accordance with the above CoP.

b) (building restrictions) Any buildings erected on the building sites identified on the plan shall be subject to the requirements of the report prepared by Engineering Geology Ltd., ref: 8447, dated 14 September 2017, and any subsequent reports. Copies of the said plan and report(s) will be held at the offices of the Council, Centreway Road, Orewa.

c) (building restrictions - stormwater control) All stormwater from buildings and paved areas on Lots 1-5 shall be collected and disposed of in accordance with the report prepared by the Stormwater Hydrology Report prepared by Cato Bolam Consultants, ref: 40320, dated 17 October 2017, unless otherwise approved by Council. The collection and disposal system shall be installed in conjunction with the erection of any buildings and shall thereafter be maintained to the specified capacity and standard in perpetuity.

Advise Notes: